

- Copy and complete these sentences:
- Voltage is
- Voltage can be called.....
- Voltage is measured.....

Answer to do now

- Voltage is a measure of the push that electrons need to move in a circuit.
- Voltage can be called potential difference (p.d).
- Voltage is measured in volts (V).

Series and parallel circuits

Learning Objective:

1. Recognise series and parallel circuits.
2. Describe the changes in the current in series and parallel circuits.
3. Describe the changes in the potential difference in series and parallel circuits.

Keywords

series circuit, parallel circuit.

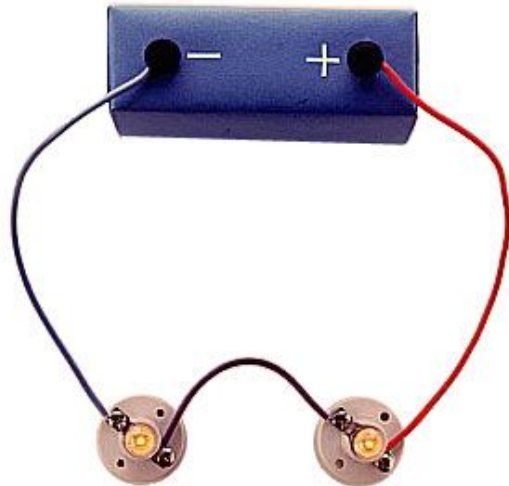


Queen
Elizabeth
School

types of circuit

There are two types of electrical circuits;

SERIES CIRCUITS

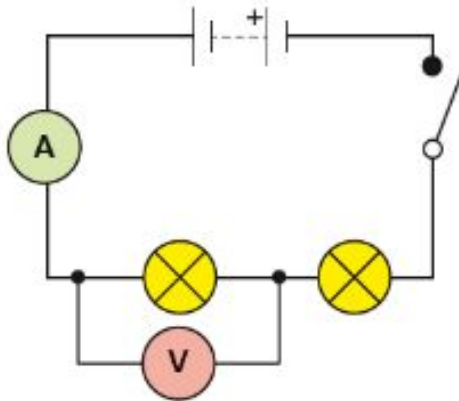


PARALLEL CIRCUITS

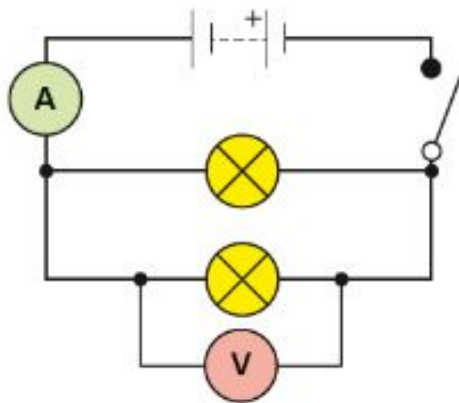


Lamps in series and parallel

Series circuit



Parallel circuit



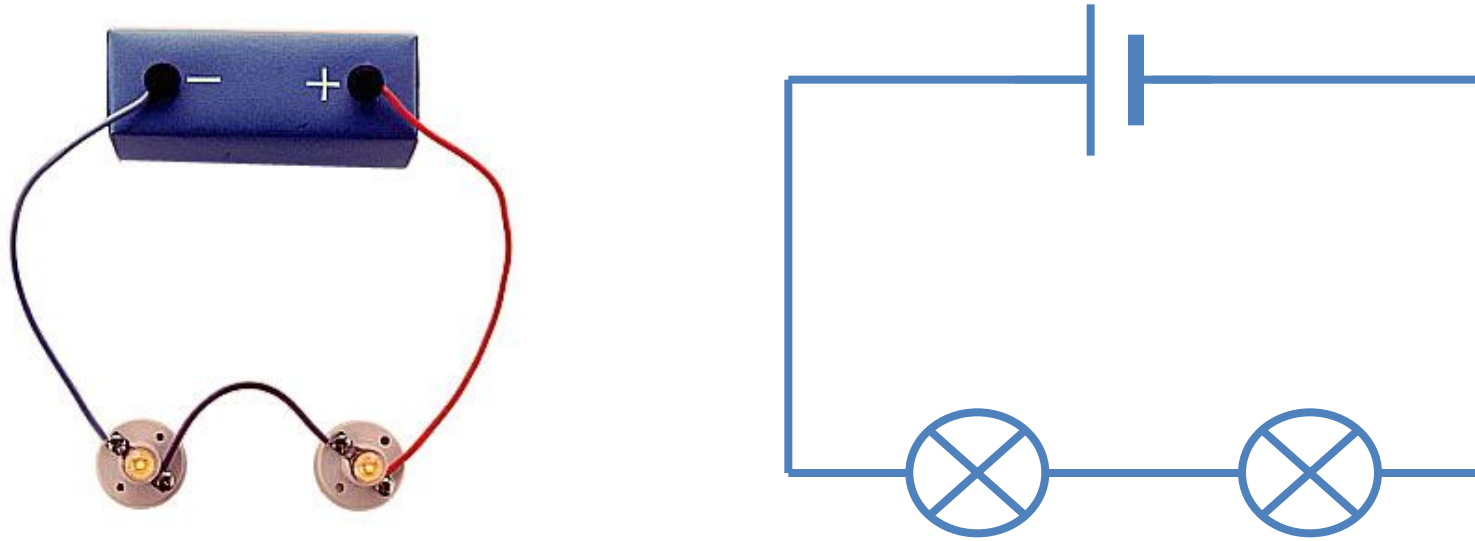
In the **series circuit** in Figure 2.6 the lamps are connected next to each other and form a single loop with the battery and the switch. In the **parallel circuit** both sides of the lamps are connected to each other – a little like the rungs of a ladder.

When they are shining, the lamps connected in parallel are brighter than lamps connected in series.

If an extra lamp is added to the series circuit, then all of the lamps in the circuit are even dimmer. If an extra lamp is added to the parallel circuit, the lamps are the same brightness as before.

Why is the brightness of the bulbs not affected in a parallel circuit compared to the series circuit?

SERIES CIRCUITS



The components are connected end-to-end, one after the other.

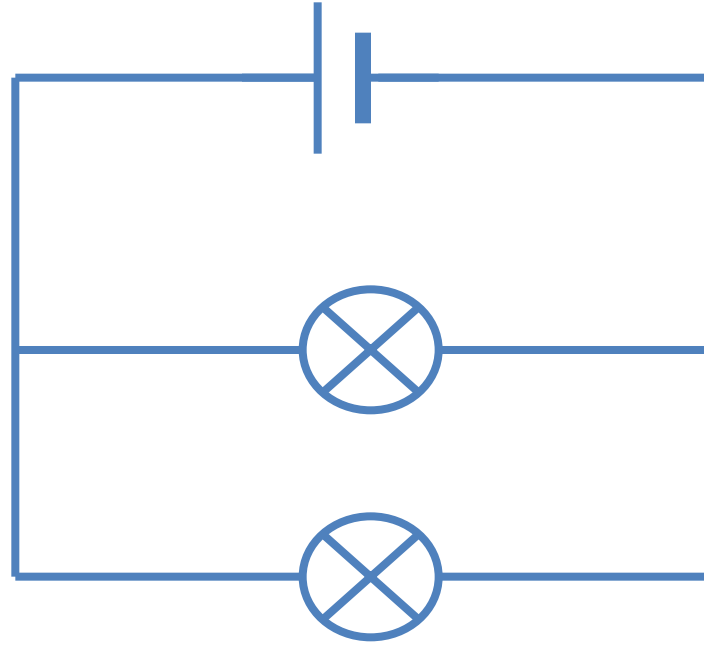
They make a simple loop for the current to flow round.

If one bulb 'blows' what happens to the whole circuit?

If one bulb
'blows' it breaks
the whole circuit
and all the bulbs
go out.



PARALLEL CIRCUITS

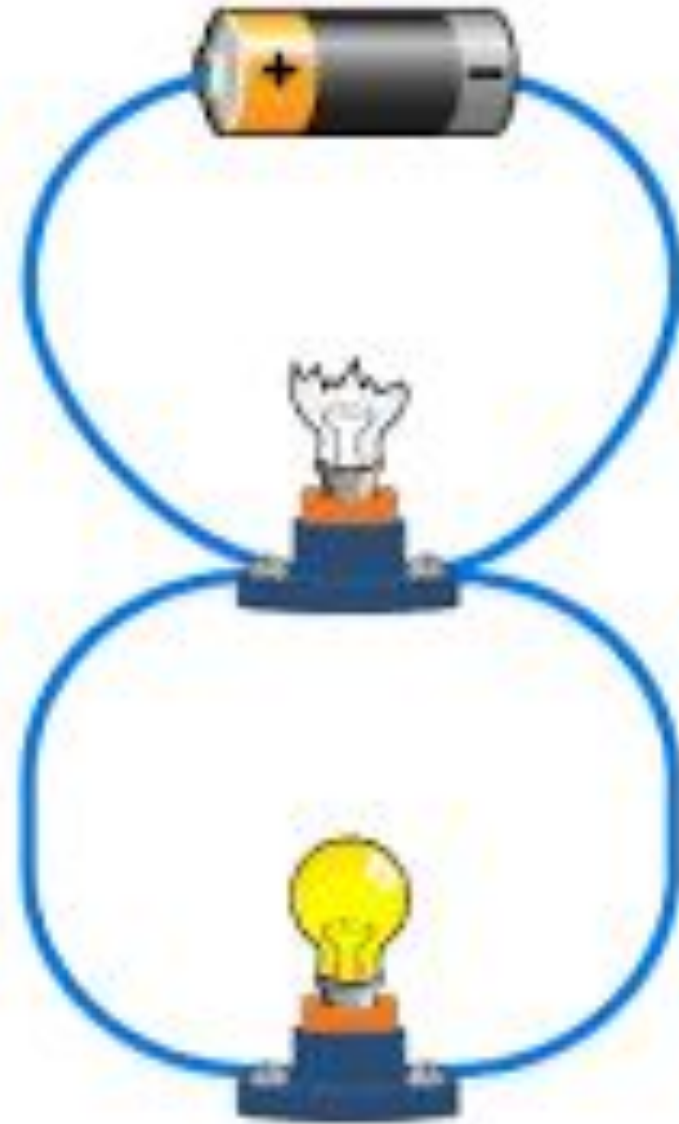


The components are connected side by side.

The current has a choice of routes (paths).

If one bulb 'blows' what happens to the whole circuit?

If one bulb
'blows' there
would still be a
complete
circuit to the
other bulb so
it stays lit.



Copy and complete the following sentences

- **Current** is a

The electrons have.....charge

- **Current** is measured in.....

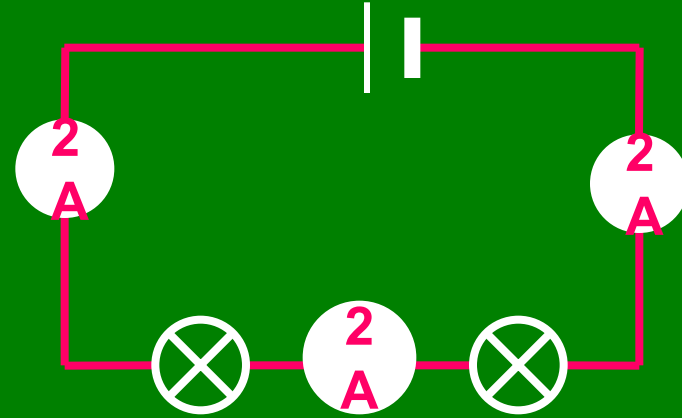
Current

- Current is a measure of how much electrons flows through a circuit.
- The electrons have negative charge .
- Current is measured in amps (A).

measuring current

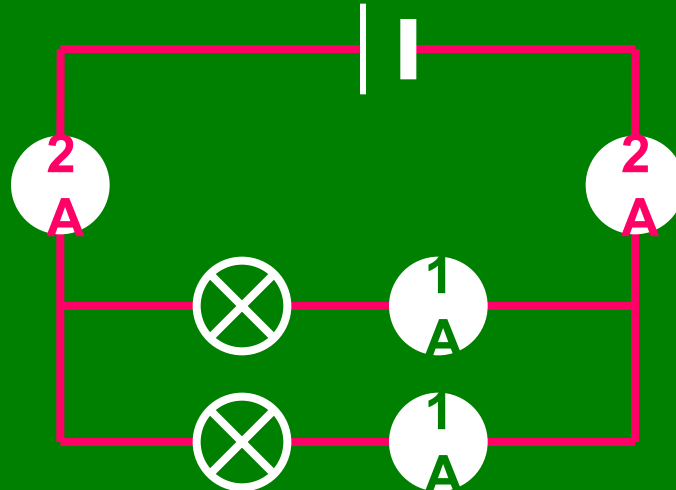
SERIES CIRCUIT

- current is the **same** at all points in the circuit.

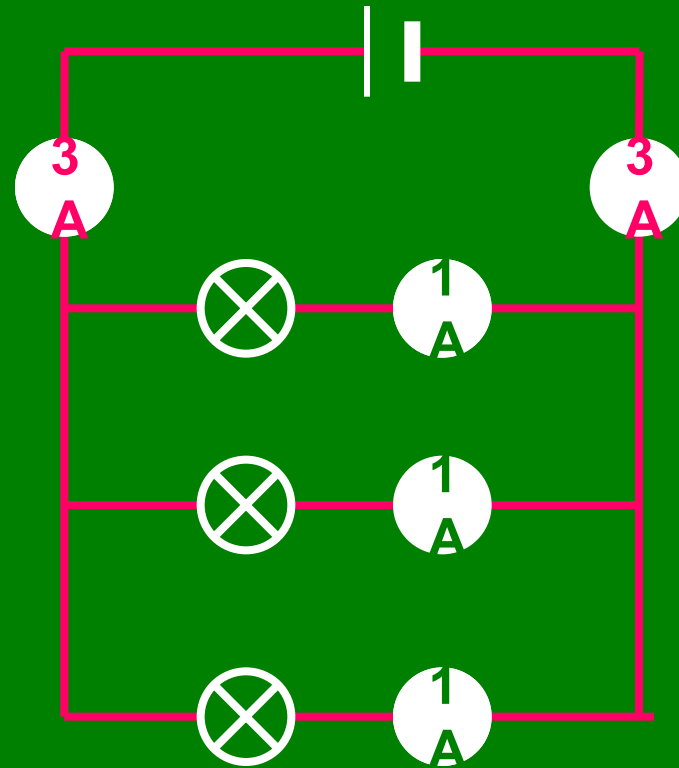
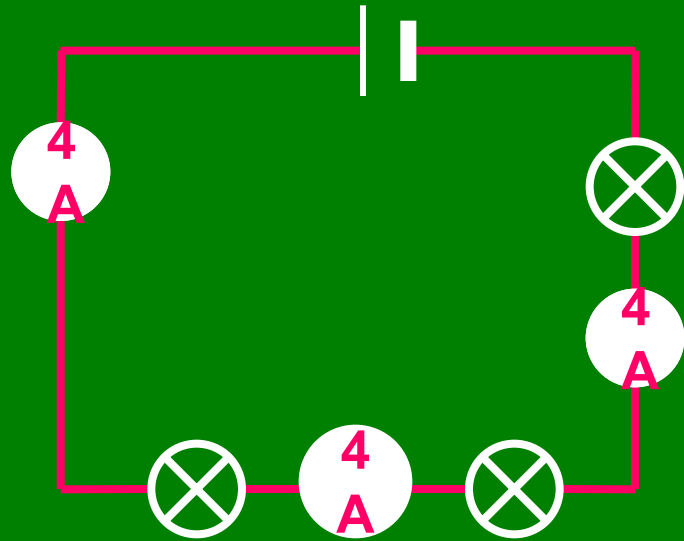


PARALLEL CIRCUIT

- current is **shared** between the components



copy the following circuits and fill in the missing ammeter readings.



measuring voltage

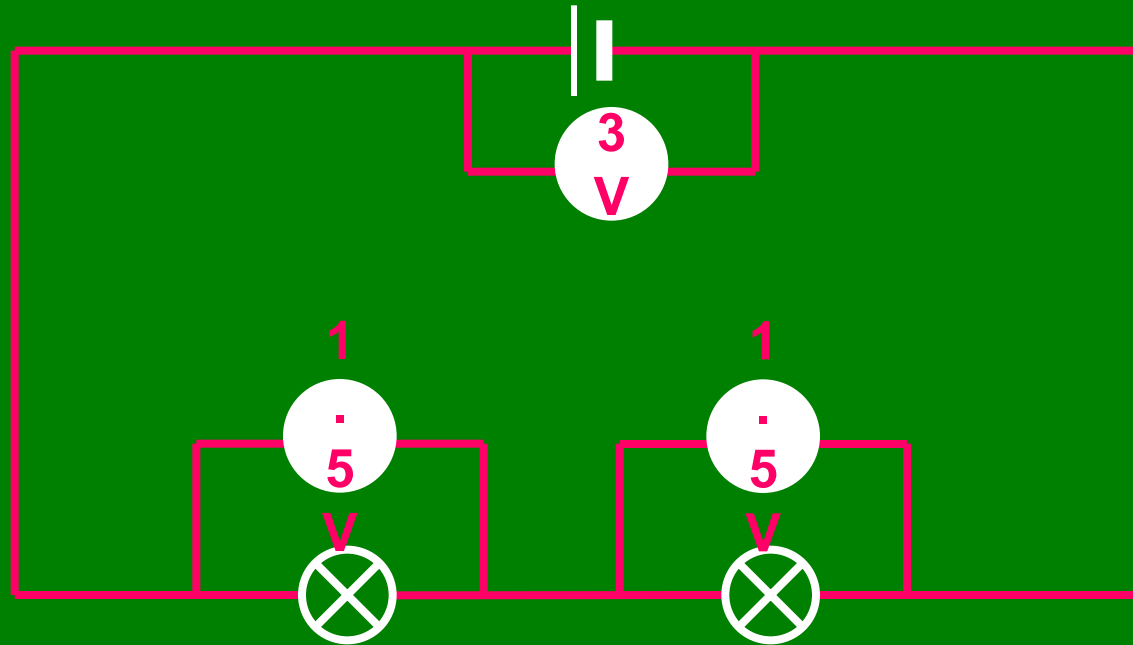
Different cells produce different voltages. The bigger the voltage supplied by the cell, the bigger the current.

Unlike an ammeter a voltmeter is connected across the components

Scientists usually use the term **Potential Difference** (pd) when they talk about voltage.

series circuit

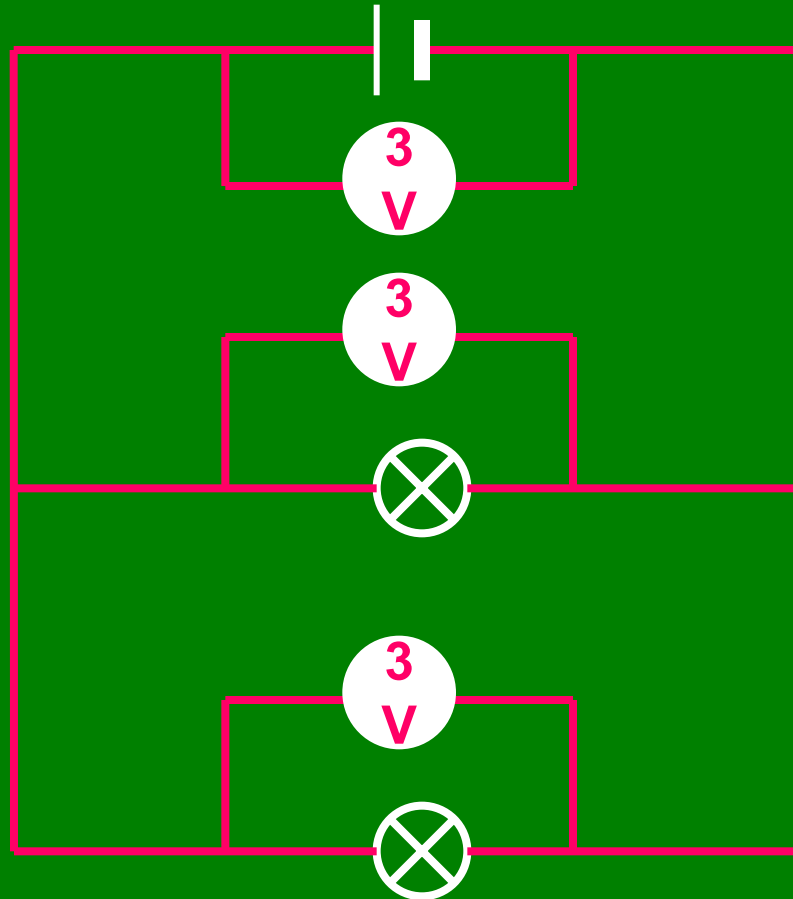
- voltage is **shared** between the components



parallel circuit

- voltage is the **same** in all parts of the circuit.

Meaning it will equal to the voltage that was given from the power source



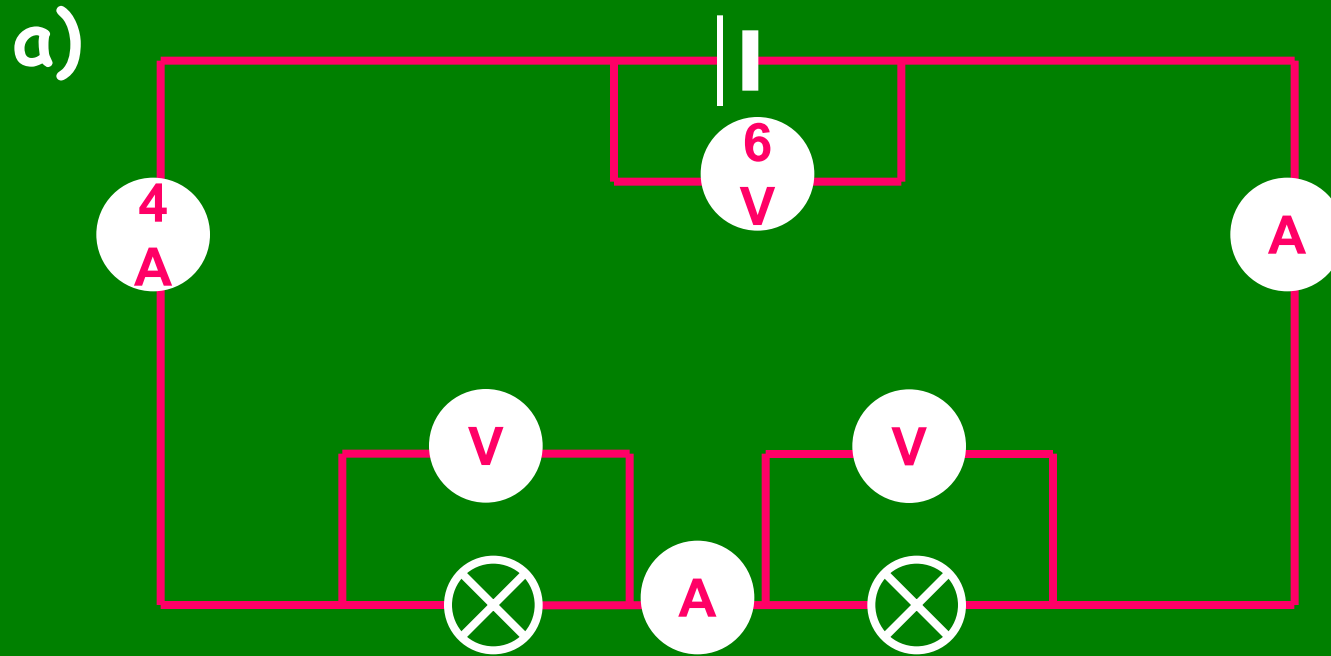
measuring current & voltage

copy the following circuits on the next two slides.

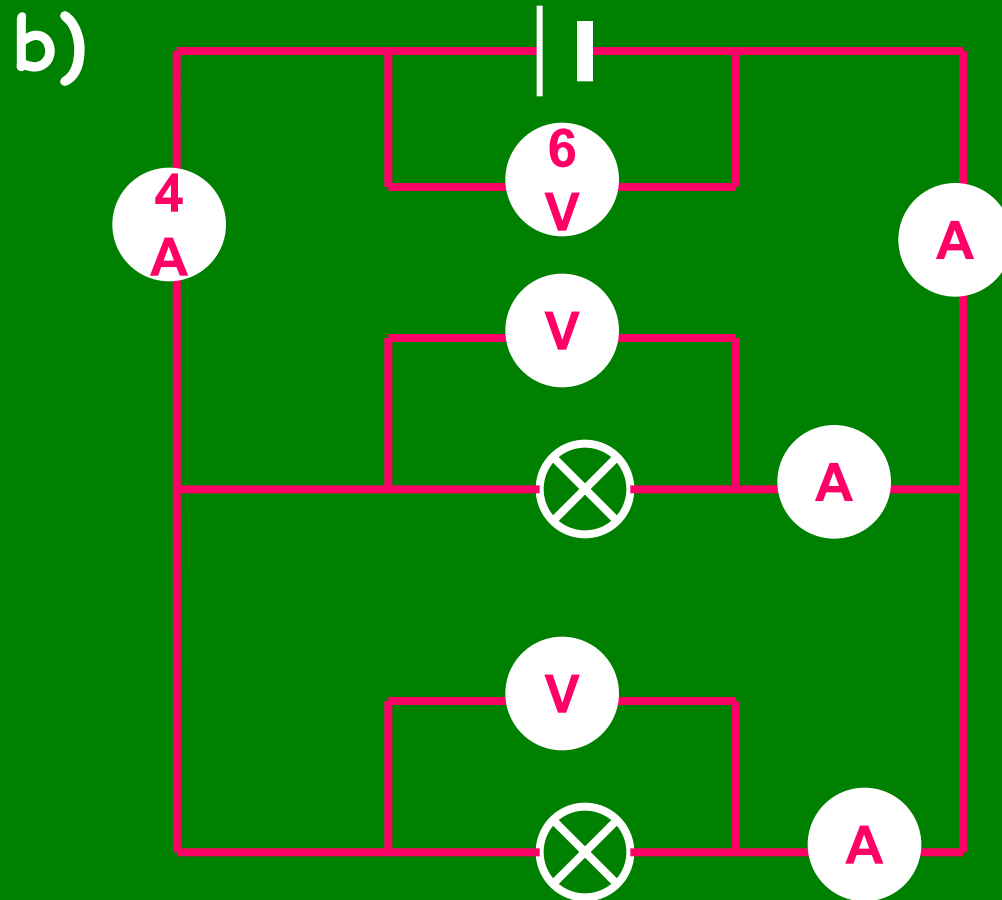
complete the missing current and voltage readings.

remember the rules for current and voltage in series and parallel circuits.

measuring current & voltage

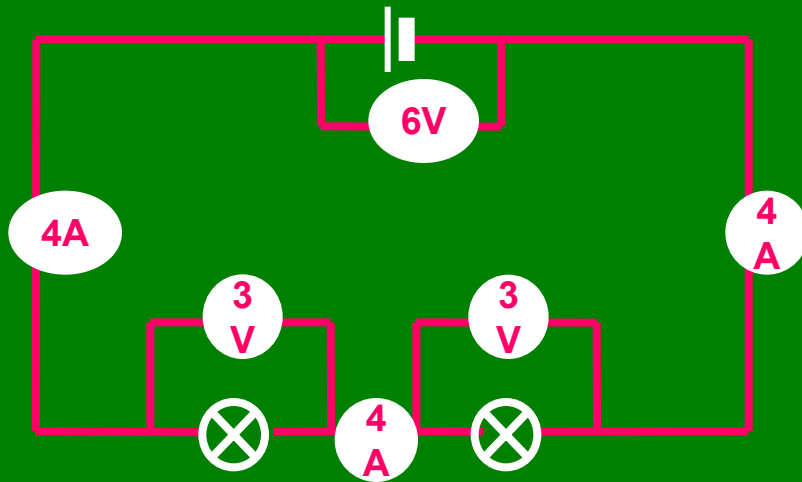


measuring current & voltage

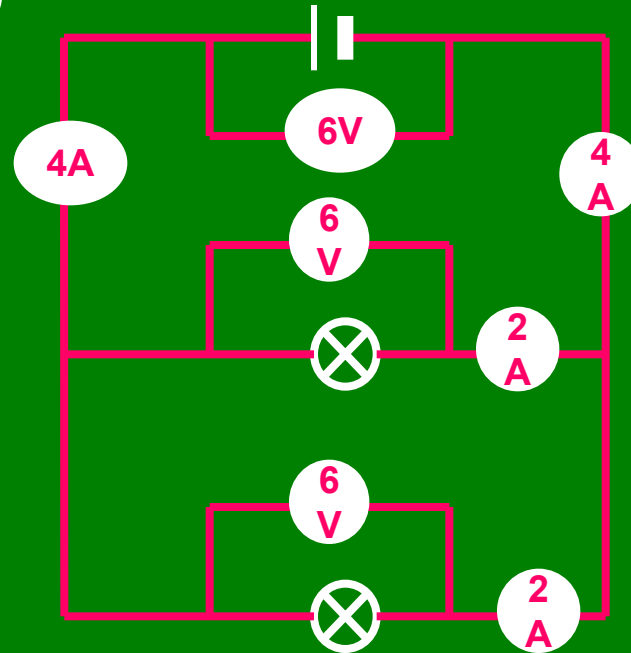


answers

a)



b)



Series vs Parallel

Series

- Current stays the same
- Potential difference (voltage) is shared between components

Parallel

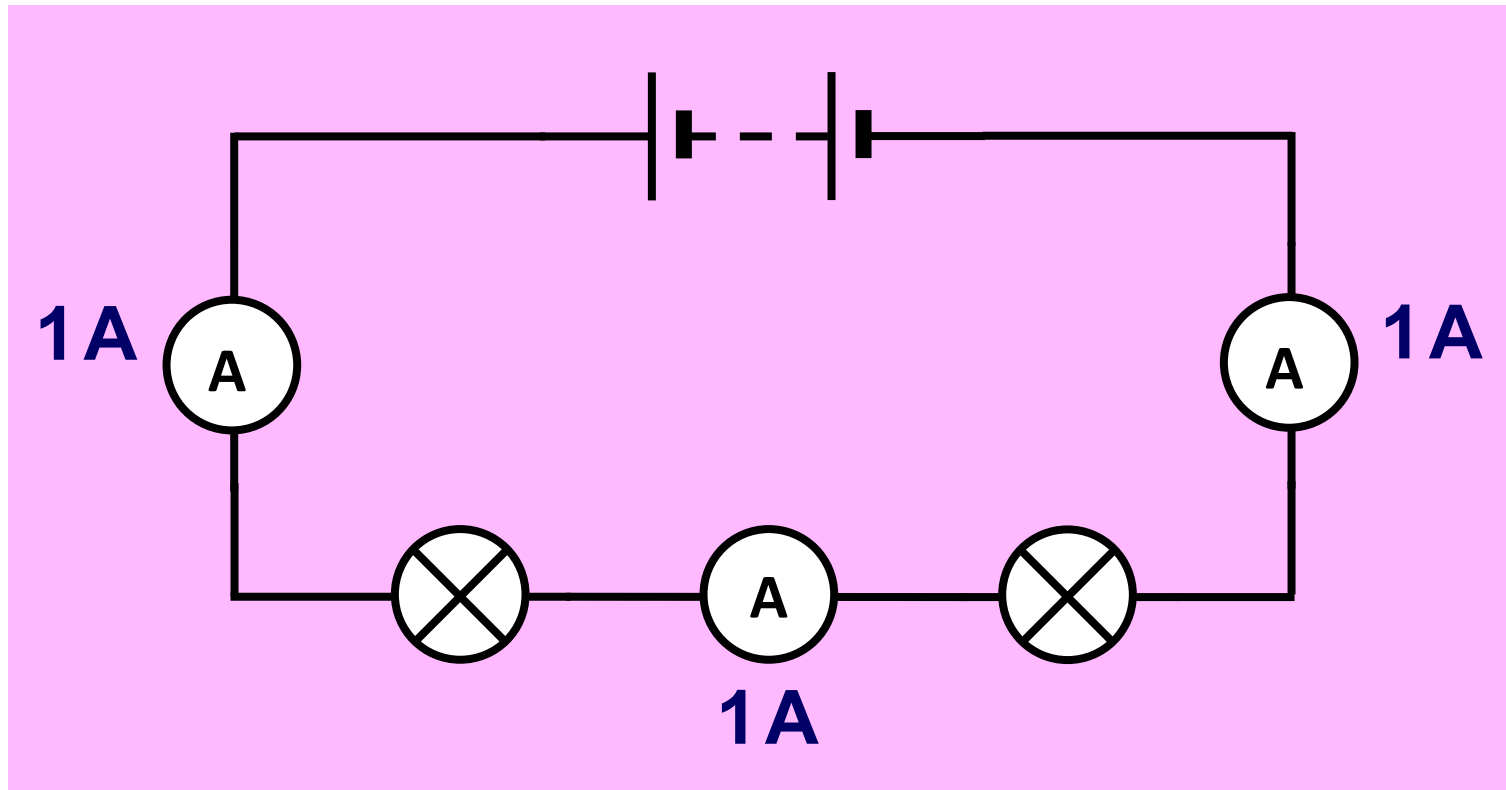
- Current splits
- Potential difference (voltage) stays the same

Resistors in series

Current and voltage in a series circuit

In a series circuit, the current flows through one continuous pathway.

So, the current is the **same** in all parts of a series circuit. The voltage is shared.



Current and voltage in a parallel circuit